SUYASH KUMAR

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EDUCATION

The University of Chicago, PhD Candidate, Astronomy and Astrophysics — September 2022 — June 2028 (expected)
PhD thesis: Physical conditions and chemical imprint of extragalactic gaseous halos — Advisor: Prof. Hsiao-Wen Chen

University of California, Los Angeles, Bachelor of Science, Physics — September 2018 — March 2022

PUBLICATIONS

- "The Cosmic Ultraviolet Baryon Survey X: Density and Pressure Profiles of Clumpy Circumgalactic Gas at $z = 0.4 - 0.7$" by **Kumar** et al., accepted to *The Astrophysical Journal*, arXiv:2610.36088 (2026)
- "Non-equilibrium ionization in the multiphase circumgalactic medium – impact on quasar absorption-line analyses" by **Kumar** and Chen, accepted to *The Open Journal of Astrophysics*, arXiv:2501.13170 (2025)
- "On the Nature of the C IV-bearing Circumgalactic Medium" by **Kumar** et al., accepted to *The Open Journal of Astrophysics*, arXiv:2408.15824 (2024)

TALKS

- "Disentangling chemical enrichment pathways in the CGM", contributed talk, CASSACA, Puerto Varas, Chile (2025)
- "Tracing CGM feedback through chemical enrichment patterns", contributed talk, Perimeter Institute, Waterloo, Canada (2025)
- "Constraining non-equilibrium ionization in the CGM", invited remote talk, Niels Bohr Institute, Copenhagen, Denmark (2025)
- "Chemical enrichment in the circumgalactic medium", invited talk, UMich, Ann Arbor, USA (2025)
- "The C IV-bearing circumgalactic medium in distant galaxies", contributed talk, IUCAA, Pune, India (2024)

EXPERIENCE

PhD Candidate, UChicago Astrophysics Department — August 2022 — Present

- Performed Bayesian modeling to infer physical properties of galactic plasmas (thermodynamics, turbulence, chemical fossil record) by combining spectral data from space and ground-based telescopes
- Built a parallelized, interactive inference engine that reduces MCMC sampling runtimes from 4+ hours (up to ≈ 80 parameters) to under 10 minutes per galaxy, enabling faster hypothesis iteration
- Led data reduction and processing for data from two cycles on the *Hubble Space Telescope*, awarded five nights of observing time for thesis research on the Magellan Telescopes in Chile
- Led software development for the Cosmic Ultraviolet Baryon Survey, implementing survey field viewer for data exploration and observation planning (check example)

AI Hackathon Lead, UChicago Data Science Institute — April 2025 — July 2025

- Led runner-up team for the UChicago AI+Science PhD hackathon, invited for post-project collaboration. Developed infrastructure for processing heterogeneous experimental datasets
- Built a message-passing Graph Neural Network to predict atomic orbital binding energies, incorporating delta learning to refine lookup-based estimates
- Designed data cleaning and parsing routines to handle inconsistencies in molecular structure representations and improve feature encoding
- Developed a scalable training pipeline on an HPC cluster for grid-based hyperparameter tuning across multiple model variants

Satellite Operations Lead, UCLA Space Physics Dept. — September 2019 — June 2022

- Led operations for two NASA CubeSats, designing real-time control measures and communication protocols, extending mission lifetime by two years
- Built and tested command-and-control software to interface with spacecraft hardware, integrating telemetry processing and fault detection routines
- Developed inference algorithms to identify sensor misalignments through matrix-based geometric modeling of magnetic field time-series data

AWARDS

- Neubauer Research Fellowship winner at UChicago, $40k award (2022 — 2027)
- Awarded travel grant from the American Astronomical Society to attend conference at IUCAA, Pune (2024)

RESEARCH ADVISING

- Tzu-Ning Yen (senior at National Tsing Hua University), co-advised with Prof. Hsiao-Wen Chen (Summer 2025 – Present)
- Lingyun Xie (freshman at Northwestern University), co-advised with Prof. Hsiao-Wen Chen (Summer 2025 – Summer 2026)

TEACHING

- Teaching assistant for four astrophysics courses designed for non-majors (UChicago, 2022 — 2023)

SKILLS

- Mathematics and statistics — Bayesian inference, Monte Carlo methods, probability theory, estimation techniques, linear algebra, time-complexity optimization, numerical methods, vector calculus, graph theory
- Programming and systems — Python, Git, Bash, parallel computing, scientific pipeline development, HPC workflows
- Machine learning tools — PyTorch, CNNs, GNNs, Gaussian processes, ensemble methods, multivariate regression